

Shireen Khan

AI and ML Engineer

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AI & Machine Learning Engineer with robust expertise in Machine Learning, Deep Learning, Natural Language Processing (NLP), and Generative AI applications. Proficient in developing AI-driven platforms, predictive analytics systems, RAG-based applications, and interactive dashboards utilizing Python, Streamlit, LangChain, and Scikit-learn. Committed to creating intelligent solutions that enhance user experience and address real-world challenges, aspiring to contribute as an AI/ML Engineer in forward-thinking, data-centric organizations.

Education

Amity University, Noida Master's in Computer applications with AI and ML, 66.00%	Aug 2024 - Present
Board of Secondary Education Rajasthan 10th, 86.00%	Jul 2017 - Apr 2018
Board of Secondary Education Rajasthan 12th Science, 85.00%	Jul 2019 - Apr 2020
Maharishi Dayanand University Bachelor's in Science (General), 72.00%	Jul 2020 - Aug 2023

Experience

GNCIPL AI and ML Intern Online - Developed Machine Learning and Deep Learning applications for intelligent automation. - Engineered AI-powered dashboards and predictive analytics systems utilizing Python and Streamlit. - Facilitated data preprocessing, feature engineering, model training, and evaluation. - Collaborated on deploying AI solutions and optimizing machine learning workflows.	Oct 2025 - Dec 2025
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Projects

AI-Powered Student Performance & Learning Assistant EdTech platform utilizing Machine Learning - Developed an AI-powered personalized learning platform to deliver customized educational content based on student learning needs. - Integrated LLM-based AI Tutor capable of answering subject-specific queries with contextual understanding. - Implemented automated notes summarization to convert lengthy study material into concise learning resources. - Built chapter-wise quiz generation functionality to enhance student engagement and knowledge assessment. - Designed an interactive dashboard for tracking student performance and learning progress. - Utilized FastAPI, Streamlit, LangChain, and Hugging Face models to create a scalable AI learning ecosystem. - Improved learning efficiency through AI-generated personalized recommendations and adaptive content delivery. GitHub	Aug 2020
Local Food Wastage Management System SQL and Python-based application - Tech Stack: Python, SQL, PostgreSQL, Streamlit - Developed a Food Wastage Management System to streamline food donation and distribution processes, reducing food waste and improving resource utilization. - Designed and implemented a responsive user interface using Streamlit for efficient data management and user interaction. - Integrated PostgreSQL with Python to build a robust database-driven application for storing and managing food inventory data. - Performed CRUD (Create, Read, Update, Delete) operations to ensure seamless data handling and real-time updates. - Executed 15+ SQL queries for data retrieval, reporting, and analytical insights to support decision-making. - Applied DBMS concepts, including database design, normalization, and relational data modeling, in a real-world application. - Conducted data analysis using Python libraries to identify food wastage trends and optimize resource allocation. - Enhanced understanding of database connectivity, query optimization, and backend development through hands-on implementation. - Improved system efficiency by integrating Python-based data processing with SQL database operations. - Developed an end-to-end solution demonstrating full-stack development skills, including database management, backend logic, and frontend design. GitHub	

Product Recommendation App

Machine Learning-based recommendation system

- Built a Product Recommendation System using TF-IDF, K-Means Clustering, and Cosine Similarity techniques.
- Conducted EDA, feature engineering, model development, and evaluation using Precision@K and Silhouette Score.
- Developed and deployed an interactive recommendation application using Streamlit.
- Improved personalized product discovery through machine learning-driven recommendations.

[GitHub](#)

May 2026 - May 2026

AskTube AI

Generative AI application utilizing open-source LLM

- Developed a Generative AI application using an open-source LLM and RAG for context-aware question answering.
- Extracted and processed YouTube video subtitles into embeddings for semantic retrieval.
- Stored embeddings in a vector database to enable efficient similarity search.
- Implemented a RAG pipeline to retrieve relevant context and generate accurate responses to user queries.
- Enhanced answer quality and relevance through embedding-based retrieval and LLM integration.

[GitHub](#)

Skills

Python

Machine Learning

Deep Learning

Artificial Intelligence

Natural Language Processing (NLP)

MySQL

Generative AI

Retrieval-Augmented Generation (RAG)

Problem Solving

TensorFlow

Large Language Models (LLMs)

API Integration

FastAPI

Flask

Model Evaluation

LLMs

Extracurricular Activity

- Internship for AI and ML - GNCIPL (Oct 2025)

Awards

- Balika Protsahan Puruskar | Balika Shiksha Foundation

Certifications

- Certificate in Advanced Excel - Amity University, Noida
- Fundamental of Computer Science - Amity University, Noida
- Certificate in Professional and Life Skills - Amity University, Noida (Dec 2024)
- RS-CIT - Vardhman Mahaveer Open University (VMOU), Kota
- Certificate Course in Leadership and Motivation in Organization - Amity University, Noida
- AI and ML - Amity University, Noida
- TensorFlow and Keras Bootcamp - OpenCV